Kalshi Market Lab

Research, mathematics and engineering audit

November 4, 2026 | America/New_York

V3 review branch - production release held

Existing public origin: https://predictir.dev. New features described here are staged locally unless an explicitly dated earlier-release check says otherwise. This report does not certify the current public deployment.

A source-backed account of hypotheses tested, models trained, code staged, costs considered and evidence still missing. This is independent project engineering; employment verification and live-money performance are outside its evidence.

Evidence class	Meaning
Staged and tested	Code exists locally and is exercised by named tests or reproducible tools; this does not mean deployed.
Measured	A dated local fixture or held-out model result, with its scope stated.
Experimental	A conditional research computation whose empirical promotion requirements remain separate.
Held / not established	No new production release, self-hosted migration, advertising optimizer or live-money return is claimed.

The notable mathematical addition is a tested composition of calibrated independent forecasts, fresh score-clock conditioning, executable depth, uncertainty-aware position decisions, a discrete risk constraint and an event-overlap graph. Established papers motivate the components; the project's own measurements evaluate this implementation.

Final local-review evidence is included. Production remains held pending the user's review.

Source base revision when rendered: e1229668926860f520db5a8fa088c7059fe2f057. Report authoring may be committed after this base.

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Earlier measurements retained: model comparisons

Pregame: lower Brier is better, same 514-game holdout



The 1.554% relative Brier point improvement is observable. Its paired whole-week uncertainty interval includes zero, so the challenger remains an explicit experimental option rather than being silently promoted. The neural candidate was trained and its non-convergence retained in the record.

In-play: separate 20-game / 60-quarter-state replay



The live replay is small and retrospective, with three correlated quarter states per game. Forty earlier calibration games estimate the score-process scale; twenty held-out games evaluate it. This does not establish a calibrated live-money advantage. The model also requires a current matched score and clock before it emits an in-play estimate.

In-play calculation

P(home win | margin L, elapsed fraction t) = Phi((L + mu*(1-t)) / (sigma*sqrt(1-t))). The independent pregame chance p0 supplies mu=sigma*Phi_inverse(p0). Sigma is estimated from historical calibration-game score increments. Current fitted sigma is 13.3579 points, with a whole-game bootstrap sensitivity range [11.4684, 15.1933].

This regulation model omits possession, strategic fouling, live lineups and non-Gaussian endgame behavior. Missing scores are not zero; missing time is not inferred from the scheduled start. Overtime and unsupported boundaries are withheld. Current game state expires after 30 seconds; research caches use five seconds for in-play estimates. Source: Stern [1]; exact capture and replay files accompany the code.

Earlier engineering measurements retained

Operation	Measured result	Interpretation
Compiled ensemble	1.056 microseconds per cached score	Local CPU inference; excludes sources, network and rendering.
Repeated fill sizing	0.295 ms including construction vs 182.996 ms	1,000 sizing queries on one synthetic 1,000-level book; cash-flow parity checked.
Book broadcast	26,699 to 1,390 bytes	Bounds displayed depth; retains full indexed source reconstruction and fresh REST execution.
Schema and fanout	See runtime-benchmark.json and RUNTIME_RESEARCH.md	Contains query plans, concurrency, backpressure and negative partition results.

The sizing curve parses and sorts the book once, builds quantity/notional/fee prefixes, then answers a quantity query by binary search. Construction is O(L log L) including sorting plus O(L) prefix work; subsequent fill queries are O(log L). This saves repeated parsing during sizing and risk enumeration. It is not a claim that the whole website became hundreds of times faster.

Audit and reproduction checklist

Inspect the frozen source hashes and split dates; reproduce native model training; verify Python/JavaScript parity; run the PostgreSQL and pgTAP checks; replay the independent cash-flow tests; review the full runtime query plans; compare hosted latency before/after region changes; and inspect a paper entry, live mark, close and confirmed settlement. Use the deployment guide for private environment setup. Credentials and private journals are excluded from this report.

Pinned PDF regeneration: create a Python environment, install tools/report-requirements.txt, then run python tools/build_research_audit.py --output /chosen/path/Research_Audit.pdf. Add --public-copy public/research/research-audit.pdf only when updating the local public artifact too. No private artifact package is required.

Current implementation and release boundary

V3 adds a neutral black interface; separate analysis/news reading modes; explicit selected-contract evidence loading; source access and pricing limits; auditable $20,000 paper accounts; cash additions/decreases; browser-scoped guest sessions; owner account synchronization with conflict preservation; and optional consent-gated analytics. Runtime, protocol and mathematical experiments are documented separately from the production data path.

The current forecasting estimates continue to require exact event/outcome matching, fresh source timestamps and explicit uncertainty. Source reputation, social votes, a smoothed context score, embedding similarity or a critic preference is never silently converted into a trading probability. A more readable long-form explanation is not stronger statistical evidence.

Self-hosting, the separate advertising optimizer, cloud provisioning, broker activation and production cutover remain held or optional. No technology name alone establishes a speed improvement, cost saving or profitable strategy. The shared-ring microbenchmark was slower than the simple same-thread queue; the report preserves that negative result.

Final verification is tracked separately from implementation. A local test pass does not prove current hosted credentials, third-party access, future source availability or production deployment.

V3 mathematics: equations, targets and accounting

The context filters, score simulation and martingale bounds below are isolated experiments requiring explicit numeric inputs and assumptions. Those experiments have no import into the forecast/sizing path, emit affects_forecast=false and supply no probability adjustment. Cash conservation instead governs staged paper-account bookkeeping; cosine/RRF supports staged evidence retrieval. These equations explain different computational roles, without certifying data or guaranteeing trading outcomes.

Scalar Kalman filter [29]

```
m_pred = m_prev; P_pred = P_prev + q * Delta_hours

K = P_pred / (P_pred + R)

m = m_pred + K * (y - m_pred)

P = (1-K)^2 * P_pred + K^2 * R
```

The assumed latent context follows a random walk and the observation error has declared variance R. Process variance q scales with elapsed hours. The final line is the Joseph-form covariance update. Source credibility is not inferred from a follower count or editorial ranking. Intervals describe uncertainty under the supplied linear/Gaussian model, not empirical coverage for market outcomes.

Three-state HMM forward recursion [30]

```
a_1 = declared prior; later a_t(j) = sum_i f_prev(i) * A(i,j)

f_t(j) = a_t(j) * Normal(y_t; mu_j, s^2) / Z_t
```

States represent assumed negative, neutral and positive context. A is an exposed transition matrix and Z_t normalizes the weights. Regular observation cadence and normalized priors/transition rows are required. The implementation uses log-domain normalization. The parameters are fixed assumptions, not learned sentiment classes, source truth or a posterior over the contract result.

Terminal regulation score-margin diffusion [1]

```
T = remaining_seconds / 60

M_T = margin + drift * T + volatility * sqrt(T) * Z

Z ~ Normal(0,1)

E[M_T] = margin + drift * T; Var[M_T] = volatility^2 * T
```

Exact terminal Gaussian sampling avoids small-step discretization error for this assumed process. Drift is points/minute and volatility is points/sqrt(minute). Regulation lead frequency is not final-game win probability because overtime exists. More Monte Carlo draws reduce sampling error, not errors from omitted possession, fouling, discrete scores, lineups or model misspecification.

Conditional maximal loss and drawdown bounds [31]

```
S_0 = 0; a > 0; E[Delta S_i | past] = 0; |Delta S_i| <= c_i

V = sum_i c_i^2; P(min_{k<=n} S_k <= -a) <= exp(-a^2/(2*V))

D = max_{0<=j<=k<=n}(S_j-S_k); V_tail(j) = sum_{i=j+1..n} c_i^2

P(D >= a) <= min(1, sum_{0<=j<n} exp(-a^2/(2*V_tail(j))))

Zero-variance tail terms contribute 0 for positive a.
```

For positive lambda, the exponential process exp(-lambda*S_k - lambda^2*sum_{i<=k}c_i^2/2) is a nonnegative supermartingale under these assumptions. A maximal crossing argument gives exp(-lambda*a+lambda^2*V/2), minimized at lambda=a/V. Applying the same argument after each possible peak and taking a union bound yields the drawdown expression. It distinguishes initial loss from peak-to-trough drawdown and can be loose or equal to one. Fees, gaps, leverage and adaptive stakes can violate its assumptions. This is not the doubling-after-losses betting strategy.

Paper account cash conservation

```
available = B0 + additions - decreases + realized_PnL - remaining_basis

released_basis = prior_basis * filled_quantity / prior_quantity

realized_close = net_close_proceeds - released_basis
```

B0 is $20,000 for migrated/new guest accounts. All entry costs include modeled fees; partial closes release proportional basis. Deposits and decreases are cash transfers, not trading profit. A decrease cannot exceed available cash or remove position funding. A confirmed binary settlement is a separate journal action, not a fictitious bid fill. These invariants are covered by deterministic cash-ledger tests.

Retained-evidence similarity and reciprocal-rank fusion [32]-[34]

```
u = embedding / L2_norm(embedding); dimensions = 384

cosine(u,v) = sum_i(u_i*v_i) / (L2_norm(u)*L2_norm(v))

RRF(document) = sum_{r in {lexical,semantic}} 1/(60+rank_r(document))

An absent candidate contributes 0; rank positions start at 1.
```

The real pinned q8 MiniLM model supplies embeddings; numeric arrays are not random or fabricated. Candidate sets are bounded to retained public evidence for the exact selected ticker. The implementation uses an experimental cosine floor of 0.15 and exact native pgvector cosine, with a tested stored-array fallback. RRF is a ranking rule, not an accuracy estimate or contract probability. Semantic similarity cannot certify truth, freshness or settlement relevance, and no HNSW performance gain is claimed.

V3 selected-contract evidence and source controls

Reviewed November 4, 2026 ET; bounded live checks were observed on November 5 UTC. No production deployment is included in this change.

Contract and source boundaries

`contractEvidence(market, {enabled, selectedTicker, redditEnabled, includeX, news, sports, music})` requires explicit enablement and an exact selected ticker before any new retrieval. Reddit remains a separate opt-in. The result supplies ranked items, readable summary and long-read views, source status, film preference sensitivity and music alternatives. It never changes a forecast, trade side or allocation.

Items retain source URLs, dates, headline/description scope, engagement metadata where supplied, source tiers and reasons for relevance. ESPN descriptions, publisher headlines and public X posts are not represented as full article reads. GDELT first-seen timestamps remain labeled as first-seen metadata. Editorial ranking weights are entity relevance 40%, source tier 25%, recency 20% and specific update terms 15%. These are review-order choices, not validated source accuracy or causal price-impact estimates. Duplicated titles are removed and publisher diversity is promoted; syndication and correlated reporting remain possible.

Recent discovery is not recent publication. Explicit historical dates in a headline or dated article path are screened out of current news even when RSS or a first-seen provider reports today's timestamp. A sports game-details/preview/recap page must match both selected teams; a page for Washington-Charlotte cannot qualify for an Atlanta-Washington trade just because it names Washington. Actual last-five and head-to-head history remains available separately as historical sports evidence. Retained search rejects explicitly historical news and each new evidence load atomically replaces the selected ticker's current public chunk snapshot, so previously rejected context does not remain searchable.

The curated source catalog records identity evidence for Shams Charania at ESPN and Fabrizio Romano at CBS Sports. This establishes reporting roles and topic scope, not a measured forecast record. Official league sources, ESPN, film trade publications and third-party music-chart commentators have different roles. The chart commentators are not official Spotify sources. No source is certified accurate based on follower count or reputation.

Access and request controls

- Public source hosts are explicitly allowlisted. Responses have byte limits, deadlines and no redirect following. Requests use a shared cache and deduplicate in-flight work. Provider 429 responses honor Retry-After and X reset timestamps; authentication failures and server failures trigger backoff.

- X uses the public account-post timeline, never private communications. At most two curated accounts are read per topic, cached for 30 minutes with a cursor. Before a cold request, the app reserves five possible post reads and account-lookup costs against durable shared daily, monthly-post and monetary allowances. At review time published pricing was $0.005 per public post and $0.010 per user resource. Defaults are 1,000 reserved post reads and $6.50 in combined read reservations per month. Provider billing can change, and reservations are conservative application controls rather than an invoice

guarantee. No live paid X requests were issued for this lane's verification.

- Reddit's official FAQ distinguishes Devvit apps running on Reddit from external Data API applications. Devvit login does not issue external OAuth credentials. Approved external OAuth access can be supplied with `REDDIT_DATA_API_APPROVED`, `REDDIT_ACCESS_TOKEN` and an identifying `REDDIT_USER_AGENT`.

- The user requested public-page collection as an alternative. The fallback checks robots rules before one bounded public JSON-page request. Reddit's live robots file returned `User-agent: *` and `Disallow: /`, so the collector returned `robots_blocked` and did not request the listing. The UI can expose a direct public search link. A CAPTCHA, 403 or 429 stops retrieval; there is no proxy rotation, auth bypass or alternate-host retry.

- A successful Reddit request examines at most 25 public link posts from one topic subreddit, cached 15 minutes across viewers. No comments or author-history calls are made. Missing/deleted content, likely automation, vote score below 3, upvote ratio below 65%, self-posts, overt opinions/promotions and duplicates are excluded. These are fallible heuristics; surviving posts are not certified human or factual. Rate and quantity limits are shared across viewers. Social reads fail closed if a durable quota/cursor store is unavailable.

Music: diagnosis and actual fallback

The active Apple endpoint `rss.marketingtools.apple.com` returned 100 US song rows with a dated feed. The legacy `rss.applemarketingtools.com` hostname failed retrieval and has been removed from the provider allowlist. The supplied backend Spotify credentials were checked in memory with one client-credentials request and one public artist search; both returned HTTP 200. No credentials or tokens were written to the repository. This demonstrates local credential capability, not production environment configuration.

An available Spotify catalog is not a Spotify artist-stream chart. Billboard song outcomes are not used as artist search queries. Spotify's search reference describes catalog search and policy restrictions. Spotify catalog references remain display-only and are excluded from AI packets and models. The July 2026 development-mode update states that quotas are shared per developer account and 429 bodies can distinguish `QUOTA_EXCEEDED`. It supersedes earlier app-count guidance; do not infer current access solely from an old migration guide.

A working additional adapter reads Kworb's US daily Spotify song chart. Its robots rules allowed public collection. The bounded parser verified 200 unique song IDs/ranks, US territory, an explicit 2026-11-03 measurement date and reported daily song-stream counts. Exact artist/song matches are exposed, with the source labeled a secondary chart mirror. This is materially useful streaming context for music questions, but it is neither all-artist US totals nor a Billboard chart. No truncated top-200 sum is presented as an artist total. Official Spotify Charts, Billboard Hot 100 and YouTube Charts remain direct review links. Billboard's robots rules disallow GPT agents, so the adapter does not disguise its identity to collect that chart.

NBA evidence

Matched sports research includes each team's last five completed games and form, the last five observed head-to-head games, recent player points/rebounds/assists/minutes and actual dated boxscore rows. Missing players or boxscores stay missing; an explicit DNP row has zero observed minutes. Player averages use observed statistics and expose coverage. Injury snapshots are current reported availability, not a reconstructed historical injury archive. For official participation status consult the dated NBA injury reports, which the league updates throughout the day.

Low-minute scenarios show actual games and win counts at the stated minutes threshold, including sample sizes and intervals. They do not identify a causal injury effect: minutes respond to coaching, foul trouble, game state, rest and injuries. Recent form, head-to-head and roster splits can overlap the model's training history; they are descriptive evidence, not additive independent signals. Retrieval is bounded to 10 recent team games, or 24 when availability comparisons are requested, with concurrency three and week-long completed-boxscore caching.

Film preference

The requested 75-80% critic emphasis is implemented as a user preference for an experimental display index, default 77.5%, with a 75/77.5/80 sensitivity view. It requires a provenance-bound exact-film critic observation with a sample count plus a same-territory demand percentile within 24 hours. Missing inputs are not fabricated, imputed or silently reweighted. No live critic/demand measurement feed was supplied, so current film coverage can be context-only.

Rotten Tomatoes defines critic-positive share separately from audience sentiment. Neither measure is ticket demand or the probability of a gross threshold. Presales, release footprint, audience scores and realized box-office gross remain separate observations. No claim that 75-80% is empirically optimal is made.

Verification

`node --test tests/evidence.test.mjs tests/workspace-research.test.mjs tests/provider-failure.test.mjs` covers opt-in boundaries, source identity and date filters, heuristic Reddit rejection, robots/access stops, shared X cost reservations, provider backoff, source-unit mismatches, DNP/missing player handling and film preference limitations. Live results and source hashes are recorded separately in `evidence-verification.json`; they demonstrate observed reads, not future availability, predictive accuracy or trading performance.

V3 conditional mathematics: full assumptions and checks

These functions are implemented in `lib/workspace/quant-context.mjs` and deliberately have no import into the evidence, forecast or sizing pipeline. Every result is experimental, states that it has not been outcome-validated, sets `affects_forecast: false`, and supplies no probability adjustment. Numeric examples below are synthetic mathematical checks, not market backtests or reported trading returns.

Context smoothing

`kalmanContext` accepts explicitly supplied, dated scores in [-1, 1]. It applies a scalar random-walk Kalman filter with process variance scaled by elapsed hours and a declared observation variance. The implementation uses the Joseph-form covariance update for numerical stability. Duplicate or out-of-order timestamps are rejected. It returns the filtered context score, gain, innovation and model-conditional variance. This follows the linear filtering framework of Kalman (1960).

`hmmContext` applies a causal forward filter to three assumed Gaussian-emission regimes: negative, neutral and positive context. Priors, transition probabilities and emission parameters are exposed. It requires regularly spaced observations and normalized transition rows. These are fixed experimental parameters, not trained sentiment labels or market-state probabilities. The forward recursion is described in Rabiner (1989).

Neither function extracts sentiment from headlines or certifies a source as factual. A smoother can make a false or duplicated signal appear stable. A future experiment needs a dated, deduplicated source archive, annotated observations, a predeclared target and chronological held-out testing. Source identity, engagement and editorial rank must not be silently treated as observation precision.

Score-margin diffusion

`simulateScoreDiffusion` draws terminal regulation score margins from an assumed Brownian process. Inputs state the current margin, regulation seconds remaining, points-per-minute drift, points-per-square-root-minute volatility, seed and simulation count. It samples the exact Gaussian terminal distribution rather than adding discretization error through small time steps. Stern (1994) provides the original sports-score Brownian framework. That paper's historical fit is not evidence that this app's chosen parameters are valid for current NBA games.

The result reports scenario lead/tie frequencies, terminal-margin quantiles and Monte Carlo sampling uncertainty. A regulation lead is a different target from a final game win because overtime exists. Continuous margins also omit possessions, discrete scoring, late-game fouling and clock tactics. The returned interval excludes these model errors; running more simulations does not fix them.

For a **synthetic** case with margin 5, 720 seconds left, drift 0.5 points/minute and volatility 2 points/sqrt(minute), the mathematical terminal mean is 11 and variance is 48. With 100,000 seeded draws the observed mean was 11.0049207151 and variance 48.0366801531. These checks verify sampling mechanics only. The hypothetical lead frequency of 0.9437 is not a real game forecast.

Conditional martingale bounds

`martingaleDrawdownBounds` requires explicit acknowledgment of a zero-conditional-drift process with deterministic absolute increment bounds `c_i`. Historical P&L cannot establish those assumptions. The exponential-supermartingale method underlying these bounds is reviewed in Howard et al. (2020).

Let `S_0 = 0`, `E[DeltaS_i | past] = 0`, `|DeltaS_i| <= c_i`, `V = sum(c_i^2)` and threshold `a > 0`. Applying the maximal exponential bound to `-S` yields `P(min_{k<=n} S_k <= -a) <= exp(-a^2/(2V))`. For each possible peak time `j`, apply the same argument to the remaining increments. A union bound gives the implemented peak-to-trough upper bound `min(1, sum_j exp(-a^2/(2 sum_{i>j} c_i^2)))`. A threshold greater than `sum(c_i)` is impossible under the supplied deterministic bounds. The result can be loose or equal to one, which is reported honestly.

An independent exhaustive check enumerates all 256 equally likely paths of an eight-step symmetric +/-1 random walk. At threshold 6, 10 paths cross an initial loss of 6 and 12 paths experience a peak-to-trough drawdown of 6. The bounds are respectively 0.1053992246 and 0.2726474761, above the exact probabilities 10/256 and 12/256. This distinguishes drawdown from initial loss and checks the math on a process satisfying the assumptions.

Fees introduce drift; leverage, gaps or adaptive stakes can violate the supplied increment bounds. These functions do not infer safety from a backtest, guarantee a trading loss limit, or implement the gambling practice of doubling stakes after losses.

Verification

Run `node --test tests/quant-context.test.mjs`. The six tests check exact Gaussian posterior updates, causal prefix behavior, timestamp constraints, HMM normalization and sign symmetry, seeded diffusion moments/monotonicity, preservation of regulation ties and exhaustive finite-path bound coverage. This is mathematical implementation verification. Promotion to any forecasting role requires a separate empirical protocol and user-reviewed scope.

V3 semantic evidence retrieval

Implemented locally for review on November 4, 2026 ET. No deployment, production migration or provider payment is included.

Actual model and verified behavior

The backend uses `@huggingface/transformers` **4.3.2** and the real Xenova/all-MiniLM-L6-v2 ONNX model. The public Hugging Face model API returned revision **751bff37182d3f1213fa05d7196b954e230abad9**, which is pinned in code. Inference uses CPU, q8 weights, mean pooling, L2 normalization and 384 dimensions. The library's feature extraction documentation describes pooled, normalized embeddings. No hash, random or fabricated embeddings are used in the application.

Actual local inference produced unit-length 384-dimensional vectors. A controlled paraphrase check yielded cosine **0.7307945** for two ankle-injury descriptions versus **-0.0231702** for an unrelated film-ticket description. This verifies basic semantic discrimination on a tiny synthetic diagnostic; it is not a held-out retrieval-quality benchmark, trading signal or factual-accuracy measure.

The actual PostgreSQL integration test passed on a disposable PostgreSQL 17 + pgvector container, including native cosine search, English full-text search, reciprocal-rank fusion, ticker isolation, parameterized malicious-query handling, idempotent migration and RLS denial for a separately granted reader. The same test passed on a separate plain PostgreSQL 17 container using stored numeric arrays and exact application cosine. Neither test connected to production.

Explicit ingestion and retained search

- `GET /api/workspace/evidence?ticker=...&reddit=0|1` is called by the explicit **Load contract news** action. It reuses matched sports/music context, retrieves selected public evidence, then indexes at most 32 public chunks. X can be disabled with `x=0`; Reddit remains a separate default-off opt-in.

- Ordinary research no longer automatically calls chart-post feeds. Headline retrieval and secondary chart work require the explicit evidence action. The assistant cannot start social ingestion.

- Only allowlisted public contract-rule fields, source headlines/descriptions/public posts and secondary chart observations enter the index. Account objects, emails, chats, portfolio balances and private journals are ignored. Spotify Web API catalog content is excluded. Credential-bearing URLs and private account/journal routes are rejected.

- `GET /api/workspace/evidence/search?ticker=...&q=...` searches only retained chunks for the exact selected ticker. It does not fetch market data, news, X or Reddit. Query text is embedded locally; it is not sent to a remote embedding inference API. First use may download the pinned public model files when they are absent from the local model cache.

- The read-only MCP tool `search_retained_evidence` exposes bounded citations to the assistant. The assistant is instructed to treat source content as untrusted evidence, preserve deterministic probability/PASS results, and ask the user to Load contract news when no retained evidence exists. Citation packets keep source URLs and observation dates.

Storage and ranking

Migration `006_public_evidence.sql` creates a dedicated RLS-protected public-evidence table and English full-text index. Its mirrored Supabase migration has identical content. If pgvector is available and installation is permitted, it adds a native 384-dimensional vector column. Search computes exact native cosine over at most 256 retained chunks for the selected ticker. It does not claim an HNSW acceleration benefit that has not been measured. pgvector documentation defines native vector storage and cosine distance.

When the extension is absent, the same real embeddings remain stored as numeric arrays and are compared with exact cosine in the application. If the model is disabled or unavailable, results are explicitly labeled `lexical_only`. With no relational backend, the app truthfully labels retention `process_memory`: up to 2,000 total chunks, lost on restart and not shared across instances. Missing migration/database access returns an availability limitation rather than claiming durable indexing.

Full-text and semantic candidate lists each contain at most 30 entries and are fused with reciprocal-rank fusion constant 60. A cosine floor of 0.15 is an experimental retrieval filter, not a calibrated confidence threshold. RRF/cosine scores cannot establish truth, news impact or an outcome probability. Syndicated or false reporting can still retrieve well; source quality and settlement matching remain separate checks.

Runtime, retention and cost limits

On the observed Windows Node process, initial model load with existing cached model files plus three texts took **415.95 ms**; a new warm query took **4.57 ms**. Total process RSS was **177.13 MiB**, including imported application/runtime libraries. This is neither incremental model memory nor a production p95/p99 measurement. An earlier first observed model-cache load plus three texts took **4.284 s**. Network cold starts and lower-resource hosts can be slower.

The model is loaded on demand, with one CPU inference at a time, at most four queued embedding requests, batches of four and a 256-token truncation limit. Text chunks are at most 900 characters with bounded overlap; no more than 32 are embedded per load. A 1,024-entry vector cache avoids repeated local inference. Defaults reserve at most **6,000 embedding texts/day** and **12,000 new retained chunks/day**, shared through the durable quota store when configured. Existing repeated content does not consume new storage reservations. These limits control application work; they are not vendor invoice guarantees.

Each explicit evidence load replaces that ticker's retained chunk set with the latest relevant public snapshot, at most 32 chunks. Public article/X/rules chunks expire within seven days. Reddit chunks expire 15 minutes after their source retrieval. Expired rows are excluded immediately and physically pruned on subsequent evidence ingestion. Explicit historical dates and different-opponent game-detail pages are excluded from current news; a newly indexed old page is not made current by its discovery timestamp. Provider first-seen times are never relabeled as verified publication dates in citations. Model files are cached outside the repository at `SEMANTIC_MODEL_CACHE_DIR`, defaulting to an operating-system temporary directory. The five observed model/tokenizer/config cache files total **23,685,697 bytes**. The Docker image declares `/tmp/market-lab-models`, creates it with node-user ownership and marks it as a volume; a read-only-root deployment must supply a writable, non-root-owned scratch/cache mount. `SEMANTIC_MODEL_ENABLED=0` disables model inference and preserves truthful lexical fallback. No paid model API is used, but CPU, resident memory, model-file transfer and database storage still consume the

host's resources.

Actual Linux Docker verification passed after `npm ci --omit=dev --ignore-scripts`: UID 1000, read-only root filesystem, 128 MiB bounded tmpfs at the cache path, real pinned q8 model download and CPU inference. Two texts returned 384-dimensional embeddings in **3,933.51 ms**, including the fresh cache download, with **228.77 MiB total process RSS**. The cache directory was mode 0700. Native inference therefore does not require granting root or enabling dependency install scripts in this reviewed image. The optional Fargate bind mount uses the task's shared ephemeral storage; it is not itself a 128 MiB quota. Production resource sizing and latency remain unmeasured.

Reproduce

Run `node --test tests/semantic.test.mjs` for deterministic boundary, scoping, fusion and route tests. To execute the actual model, set `SEMANTIC_LIVE_TEST=1`. For the real PostgreSQL test, also set `MARKET_SEMANTIC_TEST_DATABASE_URL` to a **disposable database whose name ends in** `_test`; the test refuses other database names. Do not point this test at production. The integration test applies migration 006 only inside that disposable database.

V3 paper accounts, privacy and retained records

The review branch implements `GET /api/session`, `POST /api/account/login`, `POST /api/account/logout`, `GET /api/account/profile`, and `GET`/`PUT /api/account/portfolio`. Public signup has no application route. No real account has been created and no Supabase/Auth configuration has been changed.

Guests receive a six-hour encrypted identity cookie. It contains no saved ledger. The paper UI separates guest and owner browser caches, and applies the guest data lifetime/inactivity rules independently. A guest cookie cannot be substituted for an owner cookie because encryption keys are purpose-separated. Production cookies use the `__Host-` prefix, Secure, HttpOnly, Path=/, SameSite=Strict and bounded Max-Age; no Domain attribute is set. Cookie semantics.

Owner login accepts only the configured email, delegates password verification to Supabase Auth and requires a confirmed matching user. It does not implement its own password hash database. Tokens stay inside an authenticated encrypted cookie; the backend retains only a hashed session identifier, user ID, expiry and revocation marker. Each protected action validates the retained session and obtains a verified user from Supabase Auth; near-expiry tokens refresh without extending the seven-day application session. Logout revokes the retained session and clears the browser cookie. Invalid owner cookies are cleared on the next session read. Upstream error bodies and tokens never appear in account responses. Supabase verified-user API.

Transient Auth throttling, 5xx, network failures and incomplete SDK results return 503 with Retry-After, without setting or clearing owner/guest cookies or returning a guest identity. Recovery reuses the retained owner session. Explicit invalid credentials, user mismatches, unconfirmed identities and revoked sessions can end the owner session. The client preserves local owner data on a failed session load. It checks the active browser account against the verified session before writes, refreshes identity after another tab changes accounts, and ignores stale session/portfolio/save responses belonging to an older request or a different account.

Mutating account requests require an allowed application Origin, and cross-site Fetch Metadata is rejected. The frontend proxy validates and forwards this metadata along with private cookies, preserving multiple Set-Cookie headers on the response. Account traffic is not routed through the public data fallback. Login and account quotas use keyed hashes, not email/IP values in operational logs.

Saved ledgers are owner-scoped in SQL. Writes carry an observed revision; initial insert and subsequent compare-and-swap updates each use one atomic statement. Concurrent or stale writes return HTTP 409 with the retained version, preserving both browser and account versions for an explicit UI choice. The backend validates row counts, identifiers, matching trade/position contracts, cash movements, watchlist shape and payload bounds. This remains a user-controlled simulated ledger, not a signed exchange execution history.

Financial validation replays the complete ordered trade history with dated cash adjustments. It reconciles quantity, binary-price notional, modeled fees, fill totals, net cash, proportional released basis, remaining basis and realized P/L against each retained position. It accepts valid partial closes, fractional fills and binary settlements. Costs and fees are nonnegative; a sell's net cash can be negative when the selected fee assumptions exceed proceeds, preserving an honest cash shortfall. Settlement payouts and sources must match the recorded binary result and exact exchange contract path. This arithmetic

check does not authenticate a browser-supplied market price or establish trading performance.

Every retained position needs its opening trade. First uploads with truncated history are explicitly rejected with instructions to retain/export the browser copy; the backend does not invent a missing opening balance, basis or historical P/L. The browser remains the retained copy when account synchronization fails. Importing incomplete histories would require a separately reviewed trusted checkpoint design. Same-timestamp deposits are applied before trades and decreases afterward so a financially valid ordering can be checked without inventing sub-millisecond event order.

Exact activation prerequisites

After separate approval, apply reviewed migration `004_owner_portfolios.sql` to the backend database, using the existing migration preview/apply workflow. The Supabase migration mirror has identical content. The table enables RLS and grants only the existing backend application role. Browser clients receive no direct table privileges. Keep that role's 12-connection limit and the current pool of four plus one LISTEN connection.

Private backend settings needed are `MARKET_OWNER_EMAIL`, a separate high-entropy `ACCOUNT_COOKIE_SECRET` of at least 32 characters, existing `SUPABASE_URL`, `SUPABASE_SERVICE_ROLE_KEY` and `DATABASE_URL`. The cookie handler supports the existing backend service secret as a fallback, but a separate account secret is preferred. Changing that secret signs out retained cookies. No secret belongs in a `NEXT_PUBLIC_` setting, browser bundle, committed example, report or workflow artifact.

Disable public signups in the hosted Supabase Auth project before initial activation. The application's `signup_enabled:false` does not change the project's global Auth setting. Review that setting separately; this implementation does not claim it is disabled remotely.

`node tools/setup_owner.mjs` previews the initial account operation. A later explicitly approved `--apply` also requires private `MARKET_OWNER_SETUP_PASSWORD`, 12-200 characters. It requests a confirmed user through Supabase's server-only admin API and does not send an invitation or confirmation email. It never overwrites an existing account, logs a password or configures hosted signup settings. Remove the temporary setup password from the process environment afterward. Do not execute this against real credentials until account activation is approved. Supabase admin user creation.

No password-recovery or public registration workflow is provided in this owner-only phase. Such workflows require their own email and account-security review.

Verified locally

Tests use fixture credentials and an injected Auth client, with no real Supabase account changes. Coverage includes cookie tampering/purpose separation, guest expiry, cookie flags, owner allowlisting, CSRF rejection, token refresh, transient outage recovery without cookie mutation, logout/revocation, generic provider errors, setup preview without side effects, backend/proxy cookie forwarding and service-only telemetry access. Independent financial assertions reject the fabricated-capital reproduction and tampered histories while accepting actual UI-generated partial closes, fractional/multi-level fills, high-fee exits and winning/losing settlements. Disposable PG17 tests apply migration 004 and verify concurrent initial writes, authenticated concurrent updates, stale revision conflicts, rejection before persistence and owner isolation. Production data and accounts remain unchanged.

V3 analytics configuration and privacy

Reviewed November 4, 2026, America/New_York. Google configuration is saved; the v3 application remains a local review branch and has not been deployed.

Destination and container

- GA4 property: `562468214`, web stream `16274535760`, measurement ID `G-NJN43KL0FQ`.

- Existing upgraded Tag Manager container: `GTM-5635HXLP`.

- Published container version **2**, **Public workspace measurement**. Its publication and live version were verified in the Google interface. The public `gtm.js` request returned HTTP 200 and contained the expected destination, public events and consent checks.

- The repository copy is `config/analytics/gtm-container.json`. Import with **Merge / overwrite conflicting names** into the existing workspace, review the diff, then publish. The original container was exported as a rollback copy before editing. No new Google account, paid service, permission grant or advertising account was created.

For a future release, configure `NEXT_PUBLIC_GTM_CONTAINER_ID=GTM-5635HXLP` and retain `NEXT_PUBLIC_GA_MEASUREMENT_ID=G-NJN43KL0FQ` so the revocation kill switch names the same destination. GTM takes precedence over the alternative direct GA loader; only one loader runs. Deployment remains held until application review. Do not add another pasted tag snippet to the layout.

Explicit measurement and privacy

The app loads analytics only after consent. DNT/GPC keep it off. Both published tags additionally require `analytics_storage` consent. Automatic form, browser-history, scroll, video, download and outbound-link detection are disabled. User-provided data collection, including automatic email, address and phone detection, is disabled. Google signals and ad personalization are disabled; no advertising destination is connected by this configuration.

One explicit `page_view` is sent per document; the initialization tag has `send_page_view=false`. Page URLs omit query strings, fragments and arbitrary private paths; referrers contain only their origin. The app resets optional public event fields on each event so an earlier category or market does not bleed into unrelated actions.

The event trigger accepts only `page_view`, `market_select`, `category_select`, `analysis_read_mode`, `news_read_mode`, `paper_cash_adjustment`, `paper_entry`, `paper_close` and `saved_market`. Parameters are limited to sanitized page context, public tickers/categories, summary/long-read mode and add/decrease/save/remove action labels. Amounts, balances, quantities, P/L, probabilities, account identity, passwords, tokens, chat and journals are excluded.

Reporting interface

The Reports snapshot uses the User behavior template with ten selected cards: top pages, first-user source/medium, session source/medium, new versus returning users, event count by event name, key events by event name, session key-event rate by source/medium, device category, realtime users and retention. Generic city, audience, platform-only and automated insight cards were removed from this snapshot. The saved cards were verified again after a page reload.

Four event-scoped custom dimensions are registered: **Market category** (`market_category`), **Reading mode** (`reading_mode`), **Saved market action** (`save_action`) and **Paper cash action** (`cash_action`). A ticker dimension is deliberately omitted from GA's custom-definition quota; the public ticker remains an event parameter for debugging without making every contract a report grouping.

`paper_entry` is registered as a key event counted once per session, with **no default monetary value**. It measures use of the simulator, never revenue, a purchase or an actual exchange order. The standard `purchase` key event is not emitted by the app.

The property currently has no established visitor dataset. Zero/empty cards are preserved. Custom dimensions are prospective, and the new loader is awaiting the held application release; populated production reports are not claimed. Aggregate visits cannot establish whether a visitor is a recruiter or their identity. Attribution here measures acquisition and product use; it is separate from the held advertising-optimizer project and from trading-return attribution.

Reproduction

Run `node --test tests/analytics-policy.test.mjs tests/analytics-container.test.mjs` for privacy and container regression checks. Browser review separately verified the saved Google settings. The import contains public configuration identifiers only and no Google credential. Google documentation: container IDs, manual page views, consent mode.

V3 runtime engineering and infrastructure boundaries

Status: implemented as tested modules and review tools, with account routes and bounded telemetry integrated into the pending backend code. No live deployment, infrastructure creation, external telemetry subscription, broker, self-hosted production app or advertising optimizer has been enabled. Existing stack and connection limits remain the baseline.

Measured structures and scope

`SharedEventRing` is an actual fixed-slot `SharedArrayBuffer` queue using atomic publication and consumption indices. It supports exactly one producer and one consumer, typically two workers. It is not a JavaScript object queue relabeled as lock-free, a multi-producer Disruptor, an exchange execution path or a nanosecond network-latency claim. Each publication copies bytes into a bounded slot; each read copies them out. Full returns `false` without overwriting unread events. The eventual feed integration must stop admission or reconnect/resnapshot on overflow, rather than silently losing sequence continuity. Header metadata guards shape mismatch; uint32 wraparound and concurrent ordered transfer are tested. `size` is a bounded concurrent estimate.

The synthetic local Node 26.12.1 / Windows microbenchmark over 100,000 operations measured a single-thread reference queue at about 6 ns/op, a shared 256-byte offer plus copied poll at about 409 ns/op, and a simple one-price paper match pair at about 958 ns/op in the initial run. These compare different guarantees and include no network, durable storage, scheduling or production load. The shared ring adds overhead in a same-thread benchmark. It is useful only if measured worker isolation offsets this overhead. Rerun `node tools/benchmark_engineering.mjs`; results are machine-dependent.

`PaperLimitOrderBook` is a bounded simulation with integer cent prices and integer contract quantities, price priority, FIFO at a price, maker-price fills, partial fills, GTC/IOC/FOK and owner-authorized cancellation. Self-trade prevention cancels the resting maker; FOK first checks executable external liquidity. Every result is labeled `paper_only`. This engine does not emulate exchange queue position, fees, settlement, hidden liquidity or network fills, and it is intentionally separate from the current quote-backed paper execution.

Event sourcing, CQRS and brokers

Migration `005_engineering_events.sql` adds append-only domain events, delivery receipts and per-stream projection cursors. `createEventStore({pool})` reuses the current four-connection PostgreSQL pool. It adds no connection pool or LISTEN client. Per-stream advisory transaction locks, expected versions and command IDs make concurrent appends conflict deterministically; reusing a command with different payloads fails. Projections update read models and cursors in the same transaction. An injected projection handler must perform only bounded database work on its supplied transaction client.

`publishStream` is the transactional-outbox seam for optional Kafka/Kinesis adapters. It publishes in stream order, uses a stream key and stable event ID, and records delivery only after acknowledgment. A crash between broker acknowledgment and database commit can redeliver. Consumers must deduplicate event IDs and verify stream versions. The function holds a transaction/pool slot while publishing, so use small batches, strict adapter deadlines and a single scheduled worker. It is not enabled in the request path. At current scale, PostgreSQL replay/project is sufficient; Kafka or Kinesis brings a separate cost and failure domain and remains a reviewed future adapter, not a provisioned

service. Global identity values are not treated as cross-transaction commit order.

Internal protobuf and gRPC

`proto/market_events.proto` defines the unary `marketlab.v1.Ingest/Publish` contract. `grpc-ingest.mjs` implements that small contract using native HTTP/2, protobuf varints/length-delimited fields, a five-byte gRPC frame and status trailers. It preserves uint64 sequence values as decimal strings, rejects malformed input and compressed/streaming frames, and bounds messages to 64 KiB. A real HTTP/2 request/response integration test checks the wire exchange and authentication.

The listener always binds to loopback and requires a separate internal bearer token. It does not publish a public plaintext gRPC service. The handler receives an abort signal and must honor it; client timeout does not undo an already committed event, so durable command/event deduplication remains necessary. This is a minimal unary implementation without reflection, compression, streaming, channel pooling, generated-client interoperability certification or the full gRPC runtime feature set. If the boundary becomes a cross-host production service, replace the small codec with generated stubs and a supported gRPC library, add mTLS and benchmark before enabling it. gRPC HTTP/2 protocol, protobuf encoding.

Telemetry, log filtering and operator surface

`createRuntimeTelemetry` tracks request and provider durations using bounded route/provider allowlists, rolling samples and cumulative Prometheus histogram buckets. Snapshots show p95/p99 for the last 4,096 observations per process by default. They are not fleet percentiles. A provider measurement belongs around each actual outbound attempt, including failure, while a request measurement includes the caller-visible duration; retries must not be mistaken for one network attempt. Aggregate histogram buckets in an external collector before calculating fleet percentiles. Do not average per-process p99s.

The pending backend now records request completion, Kalshi and external-source HTTP attempts, and Supabase Auth SDK calls. HTTP attempt timing ends at response headers; caller-visible request timing includes subsequent body parsing and handler work. Cache hits do not create provider attempts. PostgreSQL timing currently covers the explicit diagnostic probe, not every application query. Service-authenticated `POST /internal/diagnostics` accepts `runtime_snapshot` for the safe `{telemetry, logs}` operator snapshot or `runtime_metrics` for Prometheus text. The public frontend proxy does not expose these paths. No telemetry destination has been registered or enabled.

`createLogPipeline` accepts operational allowlisted fields, strips raw bodies, URLs, tokens, user identifiers and free-form errors, deterministically samples ordinary records at 2%, and caps 480 ordinary plus 120 warning/error records per minute. Warnings/errors bypass ordinary sampling but still have their own hard cap. Its counters expose sampling and throttling. This is bounded diagnostic evidence, not an exhaustive audit trail. Domain events, journals and required security evidence must not use a sampled diagnostic sink.

`createLogDrainHandler` verifies Vercel's raw-body HMAC-SHA1 signature using a timing-safe comparison, rejects compressed payloads, caps 1 MiB / 200-record batches, drops unstructured message bodies and suppresses exact recent duplicate deliveries with a bounded cache. Its CLI binds only to loopback; no external drain was registered. External TLS ingress, signature-secret rotation and provider-side production/source filters require a separate deployment review. A self-built receiver avoids buying a third-party logging subscription but cannot remove Vercel's drain export charges: current

published pricing is $0.50/GB with no included Pro allowance, based on uncompressed JSON. Source-side sampling controls those charges; receiver-side sampling only controls downstream volume. Vercel drain pricing, signature security.

`node tools/operator.mjs` serves a separate local operator page at `127.0.0.1:8790`. Set `OPERATOR_SNAPSHOT_PATH` to an explicitly exported JSON snapshot containing `{telemetry, logs}` and `OPERATOR_ACCESS_TOKEN` to a separate high-entropy token of at least 32 characters. The file is limited to 2 MiB, records are sanitized again, report access requires bearer authentication, and the server rejects writes, cross-origin requests and unexpected Host values. Tokens are entered in a password field, used once in memory and cleared. They never appear in URLs or browser storage. There is no trading-UI link or change-execution endpoint.

Incident proposals explain observations and hypotheses, retain safe trace IDs for correlation and say approval is required. They cannot execute changes. Confirm provider latency, query waits and event-loop delay against correlated runtime evidence before approving a fix. Increasing database connections or enabling the shared ring is not an automatic response to a slow request.

Supabase Connect choices for this application

The Connect tabs are alternative integration instructions, not six additive features to install. Current server SQL uses `pg`, certificate verification, a four-connection pool and one dedicated LISTEN connection. The Supabase JavaScript client is already available for storage and account integration.

Rank	Connect option	Use here	Cost / decision
1	Direct	Persistent backend transactions, journal, event store and LISTEN	Retain the bounded server pool; no extra library or database bill just for selecting this tab
2	Framework / client library	Supabase Auth and Storage through supported application code	Reuse the installed client; expose only publishable keys and user-scoped access in the browser
3	API Keys	Key lifecycle and role boundaries for Auth/Storage/Data API	Configuration, not a separate query layer; backend secrets never enter public frontend bundles
4	Server / build APIs	Existing Node API remains the business/auth/rate-limit boundary	Add no duplicate REST server just because the tab exists; direct browser Data API access requires reviewed RLS
5	ORM / third-party library	Consider Drizzle/Prisma only for a demonstrated schema/type-safety need	No subscription required for the library, but migration/compatibility work and connection behavior must be reviewed; do not rewrite tested `pg` transactions now
6	MCP / connect agent	Project-scoped read-only operator investigation	Agent tooling, not the production market-data path; no privileged browser key or public assistant access to administrative tools

Supabase recommends direct connections for persistent backends when IPv6 or an existing IPv4 option is available, session pooling for IPv4-only persistent clients, and transaction pooling for short-lived serverless connections. Do not buy IPv4 connectivity or switch pooling modes without a demonstrated reachability need. LISTEN requires a stable session, so it must not be moved to transaction pooling. Connection guide.

Publishable keys identify an application; user authentication and RLS determine user access. Secret/service-role keys remain backend-only. MCP access should be limited to the exact project and read-only feature groups for investigations; writes require a separate approved workflow. These changes do not connect a new MCP server or alter project keys. API key guide, MCP scope and read-only options.

Validation and remaining gates

Run the engineering, event-store, operator and log-drain Node tests. Set `MARKET_TEST_DATABASE_URL` only to a disposable PostgreSQL database to execute the integration test. It applies migration 005 and creates immutable fixture history; never point it at production. The actual PG17 fixture verifies durable idempotency, version conflicts, mutation rejection, replay, projection rollback/cursor progress and outbox retry semantics. Terraform validation/mocked plans and Docker build/start are separate checks described in `terraform/README.md`.

Pending decisions are account activation, external telemetry destination/retention, actual event-path wiring, broker adapter selection, cloud migration, external drain ingress and production deployment. No purchase or live action is implied by the implemented modules.

Current request and platform disposition

Review date: 4 November 2026. This is the review branch, not a production release. "Staged" means local code and tests exist; it does not mean the feature is live. Current managed services remain Vercel, Railway and Supabase. Self-hosting and the advertising optimizer are research only, on hold.

Product requests

Request	Review-branch status	Benefit and practical limit
Neutral black design; preserve Kalshi Market Lab	Staged	Black, Charcoal and Light are neutral gray themes. White primary controls, quiet borders and restrained financial status colors replace the blue theme. Root browser review remains the release check.
Desktop and mobile usability	Staged	Responsive market/detail navigation, horizontal category controls, larger touch controls, readable forms and account access. Automated compilation is not a substitute for phone and desktop inspection.
Contract analysis Summary / Long read	Staged	A short cost-aware screen and drivers precede optional model and source detail. Longer text does not imply stronger evidence.
Contract news Summary / Long read	Staged	Explicit "Load contract news" requests evidence for the selected ticker. Reddit starts off; source access failures remain visible.
Retained public evidence and semantic retrieval	Staged; actual CPU model and disposable PG17 checks passed	Pinned q8 MiniLM embeddings support exact-ticker FTS/cosine rank fusion. Private account/journal/chat and Spotify catalog content are excluded. Retrieval scores do not change forecasts.
Source weighting, X, Reddit and publisher reporting	Staged; access-dependent	Editorial source tiers and relevance rank help prioritize review. They are not measured accuracy, causal price impact or a trading probability. Public access blocks are respected.
Sports last five games, head-to-head, player stats and minutes scenarios	Staged	Actual dated game/box-score rows support the displays. Small and overlapping samples, missing games and low-minutes confounding remain visible. No fabricated absence adjustment.
Movie critics versus demand	Staged research context	Default critic preference is 77.5%, with 75% and 80% sensitivity displays. No index is produced without compatible, dated critic and demand observations. It does not change the forecast.

Request	Review-branch status	Benefit and practical limit
Music alternatives	Staged context	A dated secondary US Spotify song chart can supplement reporting. It does not become an artist-total or Billboard settlement feed.
$20,000 initial paper balance, including current guests	Staged	Existing positions survive migration. Cash available equals starting funds plus cash adjustments and realized P/L less remaining cost basis.
Add / decrease paper money	Staged and tested	Explicit, dated cash movements survive buys, partial closes, full closes and settlement. Decreases cannot consume committed position funds. Cash changes are separate from trading P/L.
Guest lifetime and device continuity	Staged client and server	Guest records expire at six hours; the eight-hour inactivity rule is also represented. Expired guest caches/watchlists are removed without deleting owner records. Cleanup runs when the app can execute, including the next visit after a closed browser. Browser cookies identify the guest; IP changes do not erase records.
Sign-in and saved portfolio	Staged; activation/release pending	A sign-in modal, separate guest/owner records, saved contracts, exports and bounded account synchronization exist. Conflicting ledgers preserve both versions and stop automatic overwrites. No public signup is offered.
Google Analytics / GTM and settings	Staged client; container configuration reviewed separately	GTM is preferred when its public ID is configured; direct GA is the alternative, never a second concurrent loader. Consent, DNT and GPC gate loading. Private amounts, credentials, email, chat and journal content are excluded.

UI validation: 43 focused tests passed across the reviewed UI/account suites, with TypeScript checks and a prior Next production build. They cover cash reconciliation, migration, expiry, owner-safe guest-cache removal, account isolation, empty-remote preservation, conflict handling, cross-tab identity changes, stale session responses and analytics allowlists. The final full-suite/build results and browser checks are recorded separately in the final verification section.

Supabase choices and storage

Managed Supabase currently advertises Free and Pro from $25/month. Extra compute, egress and add-ons can change the bill; this is a published starting price, not a quote for this app. Supabase pricing.

Named capability	Current decision / status	Cost and benefit
PostgreSQL	Retain existing bounded server connections	Already fits durable source history, journals and account records. Query/index measurement precedes more hardware or more connections.

Named capability	Current decision / status	Cost and benefit
Auth	Owner sign-in staged	Reuse the existing managed identity service and server-only secrets. Production account activation remains a separate step.
Storage	Existing platform capability; no bulk migration	Useful for source/artifact objects when a clear retention need exists. Storage and egress remain billable resources.
Realtime Broadcast	Current app streams remain the baseline	Broadcast is message delivery, not a historical ledger. A second stream adds load and duplicate-delivery semantics; adopt only for a measured need. Broadcast.
Realtime Presence	Not enabled	Presence tracks slow-changing shared user state, such as who is online. It is unnecessary for a single-owner paper account and is not appropriate for high-frequency prices. Presence.
Edge Functions / Functions servers	No duplicate runtime added	Existing Next/backend endpoints already own rate limits and business rules. Another runtime creates deployment and observability work.
Analytics / Iceberg buckets	Defer	Intended for large analytical datasets; currently documented as alpha. Start with bounded SQL and archived objects until analytical load is measured. Analytics buckets.
Vector buckets	Defer	Useful for a large embedding archive, not necessary for exact-ticker retrieval. Current docs mark the feature alpha; embedding generation and egress still have costs. Vector buckets.
pgvector	Native exact cosine path staged and checked on disposable PG17	The retained index is scoped to at most 256 chunks per selected ticker. The real-embedding numeric-array fallback was also tested. No HNSW speed benefit, source-truth validation or forecasting improvement is claimed.
OrioleDB / alternate storage engine	Evaluate, do not switch this app	Current Supabase material lists OrioleDB as public beta. Vendor high-concurrency benchmarks are not proof of a gain for this small connection budget. Require workload, extension and restore tests before changing engines. OrioleDB overview.
Foreign data wrappers / Wrappers / ETL	Defer	They query external data through SQL, but introduce remote latency and database coupling. FDWs do not provide RLS; keep them in private schemas if later used. Supabase FDWs.

Named capability	Current decision / status	Cost and benefit
ORM: Prisma / Drizzle	No rewrite	The libraries can help schema/type workflows. Existing tested `pg` transactions are sufficient; adding an ORM alone creates no product capability.
GUI / Studio	Existing administration surface	Useful for inspection. It is not a reason to expose the database or administration UI publicly.
Extensions	Enable only for a concrete query or operational need	Availability is not adoption. Review compatibility, privileges and upgrade support for each extension.
Direct / Framework / API Keys / Server / ORM / MCP Connect choices	Ranked in the engineering review	They are alternative connection instructions, not six paid features to install together. Reuse the current direct pool and supported client. MCP remains scoped administration tooling.
Backups / PITR / branching	Keep managed guarantees distinct	Self-hosting does not automatically reproduce managed backups, PITR or branching. Restore testing matters more than merely enabling a backup flag.
PostgreSQL 17 / requested "18.6.5"	Remain on the tested PostgreSQL 17 deployment	Upstream currently lists 18.6 and 17.11. Modern upstream server versions use two components, so "18.6.5" is not an upstream server release identifier; a provider build label must be checked separately. A major upgrade still needs host support, extension compatibility and restore tests. Versioning policy, version format.

The full Connect ranking and connection/session details are in engineering-runtime.md.

Connector and wrapper catalog: grouped by purpose

These are options to evaluate, not installed integrations. A connector needs an actual account, useful data, narrowly scoped credentials and target-specific read/write review. Querying a remote system through SQL does not remove its subscription, compute, API quota or network costs. The target list is published by the Supabase Wrappers project; availability on the exact hosted database must still be checked.

Function	Named tools / targets	Actionable choice for Market Lab
Databases and analytics	PostgreSQL FDW, MySQL, MongoDB, SQL Server, ClickHouse, BigQuery, Snowflake, DuckDB, Iceberg, Redis	Keep direct Postgres now. Add one read-only external target only when a defined historical dataset and benchmark justify it; never connect every database merely because a wrapper exists.
Object and vector storage	S3, S3 Vectors, pgvector, vector buckets	Prefer existing objects and exact entity indexes first. Review retrieval quality, retention and embedding cost before a second vector store.

Function	Named tools / targets	Actionable choice for Market Lab
Identity	Auth0, AWS Cognito, Clerk	Current Supabase Auth covers the proposed owner login. Another identity provider adds migration and account-lifecycle work.
Billing and commerce	Stripe, Paddle, Orb, Shopify	No current paper-market billing requirement. Keep disconnected; never treat payment data as public evidence.
CRM and operations	Airtable, HubSpot, Notion, Cal.com, Calendly, Slack	Useful for a separate internal workflow only when requested. They add no market-data or forecast capability to this app today.
Application / infrastructure data	Firebase, Cloudflare D1, Logflare, Infura, OpenAPI targets	Use a narrow existing API adapter first. Remote query latency and data permissions must remain visible; no credentials or foreign tables exposed in the public schema.

AI, retrieval and multimodal choices

Named tool	Status / function	Self-built-first decision and cost-benefit
pgvector hybrid search	Staged selected-ticker retained-evidence search	Actual MiniLM embeddings, FTS and exact cosine are fused with constant 60. Source dates, scope and TTLs remain binding. A tiny paraphrase diagnostic verifies mechanics; a labeled held-out retrieval evaluation is still needed before claiming quality improvement. Hybrid search.
vecs	Optional Python vector client	A convenience client for a future research pipeline; it does not require replacing the web backend or independently establish relevance.
LangChain / LlamaIndex	Optional orchestration/retrieval frameworks	The current bounded functions and source adapters are easier to inspect. Choose one framework only if it removes demonstrated repeated work; adding both creates overlapping abstractions. Supabase AI toolkit, LlamaIndex integration.
Titan	Interpreted tentatively as Amazon Titan in the Supabase integration catalog	Titan multimodal embeddings could support an actual image-retrieval task. No model or AWS account was connected, and no current text/ticker task requires this extra inference bill. If a different Titan product was intended, its identity remains unresolved. Supabase Titan example.
Roboflow	Optional computer vision platform	No image labeling/detection requirement exists in the present public-market flow. Defer until a specific, authorized visual dataset and evaluation task exist. Roboflow docs.
Mixpeek	Optional multimodal extraction/search	Potentially useful for a large video/audio/image archive. Current headlines, game rows and settlement rules do not justify a separate multimodal warehouse. Mixpeek product scope.
Fireworks AI	Optional model inference/training provider	Benchmark quality, latency and actual token cost against the current provider before adding another API. No switch or training job is enabled. Fireworks docs.
LiteLLM	Optional AI gateway/proxy	Reuse existing bounded model calls for now. A proxy adds another runtime; if later adopted for budget enforcement, current docs require a database for budgets, so a DB-less configuration must not be presented as a spending cap. LiteLLM budgets.

The preferred build sequence is existing SQL/entity matching, a small labeled evaluation set, one demonstrated retrieval improvement, then a minimal adapter. Model, embedding, storage and egress costs should be measured separately. No tool above is claimed to be new, connected or intrinsically faster.

Vercel and Railway, feature by feature

Vercel Hobby is for personal, non-commercial use; Pro is advertised at $20/month with included usage credit. Railway advertises Hobby at $5/month and Pro at $20/month, with the subscription credited toward usage. Runtime resources and applicable extras still matter. Vercel pricing, Vercel Hobby terms, Railway plans.

Capability	Vercel role here	Railway role here	Decision / cost implication
Frontend hosting / CDN	Current frontend delivery	Can host a Node frontend, but migration is not enabled	Keep the current frontend while review is pending. Moving it does not automatically lower the total bill.
API / server compute	Existing same-origin Next boundary	Persistent backend and source services	Preserve clear ownership; avoid duplicate polling and duplicate servers.
Long-lived streaming / WebSockets	Same-origin proxy / stream delivery path	Persistent backend stream handling	Verify end-to-end disconnect and retry behavior. Server uptime and egress are still resources.
Database	Use the external database integration	Can run a database with a persistent volume	Retain current Supabase/Postgres until migration and restores are reviewed.
Object storage	Platform/integration option	Bucket option	No second object store added without a measured need.
Secrets and environment settings	Frontend/server configuration	Backend configuration	Account secrets remain server-side. Adding another secrets copy increases maintenance.
Git / builds / previews / rollbacks	Existing frontend workflow	Existing backend workflow	Local commits are review inputs. Nothing in this branch authorizes a live release.
Cron / scheduled jobs	Platform capability	Platform capability	Run a single bounded job owner instead of duplicate schedules.
Scaling and replicas	Platform-managed options	Vertical and replica options	A second backend can exhaust a small database connection allowance. Measure and budget before scaling.
Custom domains / TLS	Current public frontend origin	Backend origin / ingress	Preserve same-origin account cookies. An app move also needs domain, TLS and routing work.
WAF / access controls / private networking	Existing platform controls and available plan features	Private service networking / access options	Review the concrete exposure; no new subscription or network was activated.

Capability	Vercel role here	Railway role here	Decision / cost implication
Logs / metrics / alerts	Existing platform observability	Existing runtime metrics/logs	A local sanitized operator view is staged. External retention and incident destination are undecided.
Log drains	Pro/Enterprise export option	Backend can receive/process data	A self-built receiver removes a third-party receiver subscription, not the Vercel export charge: currently $0.50/GB. Vercel Drains.
Spend controls	Usage budgets and plan controls	Resource usage and hard/soft limit controls	Use measured workload and actual invoices; no savings percentage is claimed.

Engineering and larger data tools

Named technology / request	Status	Recommendation and cost-benefit
Kubernetes / K8s	Not provisioned	Too much operational surface for one small backend today. A deployment orchestrator does not create forecasting edge.
NetBird	Research only	Private administrative access could be useful in a future self-hosted setup. Self-hosting still needs infrastructure; current quickstart specifies a public Linux VM with at least 1 CPU and 2 GB RAM. NetBird quickstart.
Caddy / Nginx	Research only	Choose one reverse proxy if self-hosting. No need for both in front of a single app. Next recommends a reverse proxy for self-hosted production. Next self-hosting.
Kong	Existing component of a full Supabase self-host stack, not deployed here	A general API gateway adds configuration and another service. It is not required for the current small app boundary.
Databricks / Snowflake / Redshift	Not connected	Separate warehouses bring compute, storage, governance and pipeline work. No measured workload presently justifies adding any of them.
Apache Iceberg	Evaluation only	An analytical table format, not a replacement for the app transaction database. Consider after a large historical archive and query benchmark exist.
Spark / Hadoop / Hive	Not installed or provisioned	Distributed batch tooling is excessive for current bounded datasets. Keep an exportable archive so this decision can change with measured scale.

Named technology / request	Status	Recommendation and cost-benefit
Provider latency p95 / p99	Isolated telemetry module staged	Bounded measurements support diagnosis. Process percentiles are not fleet percentiles; aggregate histogram buckets before reporting fleet tail latency.
Production log filtering / sampling	Isolated pipeline and signed drain receiver staged	Bounds diagnostic volume and strips private fields. Sampled diagnostic logs cannot replace an immutable audit or trading journal.
Separate operator UI / incident tracing	Local authenticated read-only tool staged	Displays safe evidence and proposes fixes. It cannot execute fixes or change the trading interface. External deployment remains undecided.
LMAX-style lock-free ingest	Actual bounded SPSC shared-memory experiment staged	The local shared-ring offer/poll benchmark was slower than the simple same-thread reference queue. Worker isolation might justify it in a different measured workload; no runtime speed improvement is claimed. It is not a multi-producer LMAX Disruptor.
Event sourcing / CQRS	PostgreSQL event store, idempotency, replay and projection seam staged	Durable, versioned events can improve auditability. Not wired into the live request path.
Kafka / Kinesis	Optional outbox adapter seam; neither broker enabled	Keep PostgreSQL event replay until actual throughput/consumer requirements justify a broker and its cost.
Custom paper limit order book	Isolated simulator staged	Tests price/time priority and order handling. It does not model real queue position, hidden liquidity or actual fills, and does not replace quote-backed paper execution.
Protobuf / gRPC	Minimal loopback unary wire implementation staged	A measured internal interface experiment. Cross-host production needs supported generated stubs, transport security and interoperability validation.
Terraform in `/terraform`	Default-disabled reference and mocked checks staged	Registry-only and full AWS profiles are separate choices. No resource has been created; a full AWS compute/database profile exceeds the additional $25/month target.
CI / Docker / ECR / deploy	Review CI and manual publisher staged; deployment dormant	Local Docker/CI validation is distinct from image publication and traffic cutover. No cloud publish or deploy is authorized by these files.
Kalman / HMM / stochastic score diffusion / Bayesian-style context experiments	Isolated conditional math staged	These operate on explicit supplied numeric observations and assumptions. No unvalidated news/sentiment score changes the released forecast, trade side or allocation.

Named technology / request	Status	Recommendation and cost-benefit
Market attribution	Observed source/price-history context remains distinct	Timestamps and price changes can show sequence, not identify a causal effect of a headline. This belongs in market research and is separate from the held advertising optimizer's campaign attribution.

See engineering-runtime.md for measurements and limitations, and terraform/README.md for the disabled infrastructure profiles and cost arithmetic.

Self-hosting assessment: on hold

Self-hosting is feasible, but it is not intrinsically free. A small first option would host the Next frontend, current backend and PostgreSQL behind one reverse proxy, with private administrative access and tested off-machine backups. Full self-hosted Supabase is a larger system and should be chosen for its Auth, Storage and other capabilities only when those capabilities are actually needed.

Supabase's current Docker guide specifies at least 4 GB RAM, 2 CPU cores and 40 GB SSD for its full stack; the recommendation is 8 GB+, 4 cores+ and 80 GB+ SSD. Running on an existing computer may avoid a new hosting subscription, while adding electricity, uptime, connectivity, backup, update and recovery responsibilities. Managed branching, backups/PITR and several platform services are not reproduced simply by launching the local CLI. Supabase self-hosting, Docker requirements.

No server, NetBird network, proxy, database migration, DNS change or cutover has been implemented for this held proposal. Compare an actual managed invoice against a tested self-host recovery plan before deciding.

Ploy and the advertising optimizer concept: on hold

Only public search results and Ploy's company-owned LinkedIn page were used. No Ploy website was opened, and no anonymous or incognito visit is claimed. The likely matching company is **Ploy / ploy-ai**, whose public LinkedIn description presents an AI growth teammate for building, optimizing and publishing marketing websites. This establishes a product category; it does not verify every feature, current pricing or measured customer outcomes. Ploy on LinkedIn.

A distinct advertising-review product could start with approved exports or read-only campaign reporting: spend, clicks, conversions, revenue, attribution window, CPA/ROAS and the matching landing page. It could identify tracking gaps, compare campaign cohorts, draft creative hypotheses and propose landing-page tests. Human review would precede publishing or spend changes. Incremental lift would need controlled tests; neither generated creative nor a higher click-through rate alone proves profitable growth. Advertising spend, model usage, platform access and hosting remain separate costs.

Google Ads retired developer tokens on 9 September 2026; access now follows the OAuth project's approval. Production access and brand verification still require review. Google Ads API access migration.

No Ploy clone, ad-account connection, campaign change or optimizer code was built. The proposal remains on hold.

Recruiter assessment and defensible project claims

This is independent project work. It is strongest as evidence of careful financial-software engineering: reproducible experiments, explicit units, source lineage, realistic cash accounting, negative results and a clear distinction between code, measurements and deployment. No employment responsibilities, professional transaction work or live-money results are established by this audit.

Hiring context	Evidence strength and limit
Software / fintech engineering	Strong project discussion material: shared ingestion, bounded retries, schema/transaction tests, account conflict handling and clear public/private boundaries. A reviewer should still inspect ownership, source code, test runs and operational behavior.
Data / analytics / applied ML	Credible experimental design: chronological partitions, an independent baseline, proper scores, calibration and recorded model failures. The 1.554% Brier point improvement has uncertainty crossing zero; it is not a validated general lift.
Quant research	Useful demonstrations of cost-aware decisions and mathematical implementation. Missing prospective calibration, realistic strategy evaluation and transaction-cost/outcome evidence prevent an alpha or profitable-strategy claim.
Low-latency / HFT systems	The SPSC ring, protobuf/gRPC and book simulator are bounded experiments. Same-thread local benchmarks exclude networks and exchange execution. They do not establish HRT-scale systems, production latency or equivalence to a trading firm's infrastructure.

Claim wording that the evidence supports

Built a source-backed prediction-market research and paper-trading application with explicit settlement rules, dated evidence, fee/depth-aware execution, shared streaming and auditable paper cash movements.

Trained and evaluated NBA model candidates on chronological historical partitions; retained experimental promotion gates and exported the selected model with native-inference parity.

Implemented and tested conditional context filters, regulation-score simulations and martingale bounds with declared assumptions; kept unvalidated context out of the forecast and allocation path.

Do not claim newly discovered mathematics, proven source accuracy, guaranteed fills, realized live-money profit or equivalence to HRT, Two Sigma or Citadel. Their primary materials describe broad research/engineering practices and do not certify this project [7]-[11].

Final local verification and unresolved release gates

Status: final-local-review

This report describes the local review branch. The V3 application has not been deployed. Earlier published-release checks are retained separately and do not certify V3 production behavior.

Completed checks

- Forty-three focused UI/account tests passed, covering cash conservation, legacy $20,000 migration, guest expiry, owner-safe expired guest-cache removal, cross-tab identity changes, stale session responses, owner synchronization conflicts and analytics allowlists. TypeScript checks passed after those fixes.

- The final aggregate run passed **286 tests**: **276 Node tests plus 10 TypeScript tests**, with **zero failures and two intentional default opt-in skips**. TypeScript checks and the production build passed with both public GTM/GA IDs configured. Six live opt-in checks passed separately; those runs are reported separately rather than added to the aggregate as unique tests.

- Actual MiniLM CPU inference and disposable PostgreSQL 17 checks passed for native pgvector/FTS/RRF, numeric-array cosine fallback, ticker isolation and RLS. A Linux read-only-root, non-root Docker run downloaded the pinned q8 model and produced real 384-dimensional embeddings. Exact captures and scope are in the semantic verification record.

- Backend account tests exercise canonical financial replay, owner isolation, concurrent revision conflicts, cookie tampering, CSRF rejection, token refresh and logout. Truncated first uploads are rejected explicitly while the browser copy is retained. Transient Auth failures preserve owner cookies and return an unavailable response rather than switching the user to a guest.

- Local desktop development-browser review observed the neutral black workspace, changing contract titles with old quotes cleared, explicit news loading, real retained-evidence search responses and cash additions/decreases of $100 reconciling to the starting $20,000.

- A clean navigation then recorded one **PHI YES** paper contract for **$0.93 including modeled fees**, closed it against fresh bids for **$0.90 net proceeds**, and produced **-$0.03 incremental realized paper P/L** with **zero open quantity**. The exact contract was KXNBAGAME-26NOV04PHIDET-PHI. Existing simulated history was preserved. This is a bookkeeping/execution-path check, not a return backtest or a real exchange order.

- Mobile layouts have CSS and logic checks. A dependable phone-sized browser viewport could not be established in the available visual tools; no claim of complete phone visual testing is made.

Google configuration saved separately

GTM container GTM-5635HXLP version 2, Public workspace measurement, was published and checked in the Google interface. Its public script returned HTTP 200. Both tags require analytics_storage consent, and the app retains a single-loader path with privacy-safe explicit events.

Four custom dimensions were verified again in a fresh Google interface tab. The ten-card Reports snapshot persisted after reload. paper_entry is a once-per-session key event with no default monetary

value. Automatic forms/history/user-provided-data collection and advertising features are disabled in the reviewed configuration.

The property cards remain empty/zero and the new application loader remains held with the application release. Configuration persistence is verified; actual consented production visitor collection and recruiter identity are not established.

Remaining release boundary

A local production-server launch was rejected by automatic review with the generic message "blocked by policy" and no detailed reason. That launch was not retried. The production build passed; actual browser checks use the already authorized loopback development preview. Production-preview browser verification is not claimed.

No new Vercel/Railway application deployment, Supabase production migration, real account activation, AWS resource creation, self-hosted cutover or advertising optimizer is implied by the completed local checks.

Guest expiry is enforced while the app can execute and on the next active visit. A closed browser cannot run physical deletion at the six-hour wall-clock instant. Cleanup preserves active owner records.

Earlier release audit: retained historical record

The following section preserves prior model, engineering and hosted-check observations. It is not a statement that V3 was deployed or that the current public site has passed these checks. Earlier theme labels, platform prices, model-catalog observations and region/deployment checks retain their original scope; current decisions above supersede them. No historical benchmark is extrapolated into a present profitability or runtime claim.

Reviewed November 4, 2026 (America/New_York). The full requested feature list is tracked below. Implemented features use this repository and the existing Vercel, Railway and Supabase accounts. No new paid integration was purchased. A local benchmark is not an end-to-end latency or trading-return result.

Forecasts, position decisions and portfolio mathematics

The default NBA probability remains the regularized team-strength model. A separately selectable challenger trains ridge logistic, depth-two XGBoost and a 16-by-8 neural network on actual historical scores. A selection partition chooses the classifier/blend, a separate partition fits beta calibration, and a final chronological partition evaluates it. The selected composition is 75% team strength and 25% lag-feature logistic before beta calibration. The neural network's non-convergence is recorded rather than silently discarded.

Across 514 held-out games, Brier error decreases from 0.2025336534 to 0.1993854751, a 1.554% relative point improvement. Log loss decreases from 0.5935887721 to 0.5860446437. The paired weekly 95% improvement interval is approximately [-0.0011701, 0.0075196]. This promising experiment is usable through the experimental model setting; its formal promotion gate remains false. The held-out results concern outcome forecasts, not historical entries, fills or realized returns. See `NBA_RESEARCH.md`, `nba-challenger-validation.json` and `nba-inference-benchmark.json`.

The experimental in-play NBA model conditions on an actual current score and clock, following Stern's score-difference diffusion. Its prior comes from an independent pregame model. Its score-variance scale is fitted from historical calibration games, not invented. It refuses unsupported overtime and missing/stale live state. `NBA_INPLAY.md` and `nba-inplay-validation.json` record the historical state replay, parameter estimation and limitations.

The project-specific mathematical composition is an auditable combination of established methods, not a claim to a newly proved theorem. The new elements in this application are their integration with exact event mapping, dated model residuals, current executable depth, explicit uncertainty and risk constraints, and the known-event exposure graph.

For a selected side with independent chance p, range [l,u], quantity q and all-in entry cost C(q), expected settlement profit is p*q-C(q). Its sensitivity range is [l*q-C(q),u*q-C(q)]. If B is the entered budget, winning and losing ending values are B-C(q)+q and B-C(q). The chance of reaching 2B is calculated from these two actual branches; if neither branch reaches the target, it is zero. This is a settlement calculation, not a fabricated price-path forecast.

Budget scenarios spend only what the current book and fees permit. A separate discrete Kelly optimizer maximizes the lower-bound expected log growth subject to a drawdown-moment condition. With

lambda=log(beta)/log(alpha), it checks l*(W_win/B)^(-lambda)+(1-l)*(W_lose/B)^(-lambda)<=1. The displayed defaults alpha=0.8 and beta=0.1 are explicit risk-policy assumptions. A repeated-wealth guarantee would require correct stationary probabilities and repeated independent opportunities; those conditions are not claimed for this app.

The live paper tracker compares q*p with current net bid proceeds R(q). HOLD requires the lower value q*l to exceed R(q) plus a visible action buffer. EXIT requires even q*u to fall below current proceeds minus the buffer. Overlap produces REVIEW. TRIM applies an explicitly entered position-loss limit. A falling quote alone is not evidence that the original thesis failed. Partial depth, stale quotes or missing model evidence withhold full-position P/L or actions. Confirmed binary settlements are recorded separately from bid closes. Multiple open lots share one proportional allocation of their common bid book, with separately rounded lot fees; the portfolio cannot reuse the best bid or count unavailable liquidity twice. Wrong-contract packets, failed refreshes and pregame estimates after event start cannot issue current model actions.

Validated rankings require a positive lower-bound edge after the budget-sized fill and buffer. Experimental rankings allow positive point-estimate edges while showing the lower-bound risk and failed model-promotion gate. Every row uses the same entered budget as an alternative scenario; the app does not pretend that each row creates additional portfolio cash. The event-overlap graph prevents duplicate ranked exposure to a known common event. Unknown cross-event correlations remain unknown.

Complete feature disposition

Requested item	Implemented or evaluated result	Cost and benefit basis
Enums	Native PostgreSQL job-transition state enum; application model/theme/action inputs are bounded allowed sets.	Rejects invalid state transitions without another service.
Safe deletion, object drops and cascade deletes	Forward migration preview, explicit apply, hash drift checks, advisory locking and destructive-DDL rejection; immutable evidence/journal guards; no production deletion or drop was performed.	Prevents accidental destructive operations and preserves provenance. Cascades are not introduced for financial evidence.
Edge Functions	Existing Vercel API functions provide the same-origin execution layer. Supabase Edge Functions were priced and evaluated as an additional hop.	A second execution service is not provisioned where the same operation is already implemented in code. Region colocation is measured separately.
Debugging and general logger	Structured request completion, duration, status, trace ID and stream health; credential/body/query redaction.	Operational evidence without paying for an additional log product or logging private payloads.
W3C Trace Context	Validated traceparent propagation and new spans; controlled tracestate handling; untrusted baggage is dropped.	Correlates frontend/backend requests while limiting metadata leakage.

Requested item	Implemented or evaluated result	Cost and benefit basis
Realtime Protocol and low-latency Broadcast	Shared native exchange WebSocket, correct control-message sequencing, subscription cleanup, bounded snapshots and one multiplexed portfolio stream.	One upstream feed fans out to multiple clients; no duplicate managed feed subscription.
Presence	Same-origin ledger coordination and browser-tab update mechanisms provide the needed local synchronization. Public online-user tracking is not added.	Supports this personal paper workflow without sending private activity to a presence provider.
Postgres Changes and triggers	Semantic job-state transitions generate an append-only typed outbox and lightweight notifications; listener wakeups complement polling recovery.	Notifications follow changes, not every quote or health request. No public private-table replication channel is enabled.
Scalability benchmarks	Reproducible model, sizing, stream fanout, database and partition/projection experiments; bounded capacities and backpressure.	Measured workload fixtures, not an unmeasured claim of HRT-scale infrastructure.
Trade tracking	Live executable marks, cost basis, fees, realized/unrealized P/L, hold/trim/exit/review decisions and explicit confirmed settlement.	Works from actual public data; no real-order route.
Recommended trades and price ranges	Budget input, independent model chances, payoff branches, target probability, cost ceiling, risk size and an event-overlap graph.	Mathematics runs in application code; language-model tokens are not used to calculate prices or odds.
Cheapest MCP	Existing read-only MCP server plus the current low-cost gateway model; selection follows the website's chosen research model.	Current catalog pricing and subscription-credit eligibility are documented; no extra MCP subscription.
GraphQL	A projected/nested read comparison is benchmarked; active search returns nested source provenance in one SQL query.	No extra GraphQL gateway or parser is purchased merely to duplicate a bounded query.
Graph algorithms and graph data engineering	Known-event exposure graph, deduplication and relational source/event/contract/observation lineage.	Captures known dependencies without purchasing a graph database or inventing unknown correlations.
Globally distributed Realtime	Vercel CDN distribution and regional backend deployment are assessed; the maintained market feed has one authoritative region.	A second paid regional feed is not claimed as deployed. Colocation and fanout benchmarks quantify the current tradeoff.

Requested item	Implemented or evaluated result	Cost and benefit basis
Flutter	Web interface supports desktop/mobile layouts without a second native codebase.	Flutter remains a costed client-port option; it does not improve the current browser runtime by itself.
Local Development and CLI v3	Pinned public Node/Python manifests, reproducible frozen data, disposable PostgreSQL tests and migration preview/apply tooling.	Stable commands remain the release path; experimental CLI commands are inspected before use.
Experimental supabase stack commands	Supabase CLI 3.0.0 and its experimental help were inspected; the named stack subcommand was not listed in that installed inventory.	No unverified experimental stack deployment is claimed. Stable local start and migration/test commands remain independently runnable.
pgTAP	Actual SQL assertions validate schema policies, transition notifications and immutable-record behavior.	Tests run on a disposable database; production data is not deleted to test deletion guards.
pg-delta and migra	Schema drift and destructive DDL risks are covered by migration hashes and reviewed additive SQL.	Blind generated diffs are not applied to the hosted financial-evidence schema.
Seeding historical data	Frozen real NBA game captures and source hashes reproduce training; synthetic test fixtures are labeled separately.	Real historical scores are not relabeled as prospective predictions or trade fills.
Full Text Search	Generated text-search document, GIN index, bounded parameterized websearch query, latest-per-contract results and source joins.	Active retained-history search; selecting a result requests fresh contract context.
Webhooks	Internal outbox/notification wakeups and same-origin browser update events serve current change propagation.	No public webhook secret or paid webhook relay is needed for this internal pipeline.
Partitioning	Local flat-versus-time-partitioned query/pruning experiment and growth thresholds.	Partitioning existing small tables is not claimed as deployed; benchmark evidence guides a later additive migration.
pgloader import	The current typed JSON/SQL ingestion validates provenance and supports batch inserts.	pgloader is reserved for an actual external database import; there is no external legacy database to migrate.
XGBoost, neural networks and ML	Actual trained ridge, XGBoost and neural candidates, chronological comparisons, calibration, compiled inference and parity tests.	Training runs locally; production does not need Python, a GPU instance or a managed ML endpoint.

Requested item	Implemented or evaluated result	Cost and benefit basis
Joins and nested tables	Parameterized source joins produce nested provenance for retained observations.	One database query replaces a client-side N+1 source lookup.
jsonb and arrays	Typed financial columns remain separate from structured metadata; JSON batch ingestion and quality-flag arrays have explicit grain checks.	Avoids an indiscriminate JSONB/GIN index over every field.
Indexes, B-tree, binary tree, BRIN and GIN	Workload-specific ordered lookup, compact time-range index and full-text index; prefix fill curves and indexed books serve numerical queries.	EXPLAIN and synthetic-scale benchmarks determine selection. No blanket indexing or routine production REINDEX.
Index advisor and database statistics	Read-only catalog/statistics, index-use and query-plan inspection plus reproducible local tests.	Does not assume that a recently created index's low usage proves it is unnecessary.
RLS	Application schema and new tables remain isolated from anonymous roles; explicit backend grants and owner-token checks continue.	No private journal data is added to public realtime, search or AI tools.
WHERE, ORDER BY, LIMIT and EXPLAIN	Bounded queries, latest-row selection, proper ordering and explicit source/time predicates.	Plans are inspected at representative synthetic scale and compared with actual small production statistics.
npm, pnpm, yarn and Homebrew	npm's pinned lockfile is the supported release install; alternative package managers and Windows package installation are documented options.	Multiple conflicting lockfiles or macOS Homebrew do not improve this Windows release's reproducibility.
Black futuristic UI and layout options	Graphite, Midnight and Light; comfortable/compact density and layout settings; visible model mode and readable controls.	Reuses the app's design language and requires no paid design service.
Top public traders, all-time/week/four months	Provenance-checked public closed-trade analysis tool computes observed-sample top-30 windows, fees, concentration and entry-dollar return.	Official public leaderboard retrieval was throttled; a complete verified top-30 history and trader motives are not invented.

Public trader research and source limits

Kalshi's official help describes an opt-in leaderboard. Public leaderboard pages show projected profit rankings, which differ from verified account-wide realized return. The public trade API does not establish every participant's identity, capital, hedges or private positions. The direct public leaderboard request returned HTTP 429 during this audit; the access limit was respected. No private profiles, messages or contacts were accessed.

The supplied-data tool `node tools/analyze_public_traders.mjs public-closed-trades.json analysis.json` accepts only explicitly public, source-linked, deduplicated closed-trade cash flows. It calculates all observed history, trailing seven days and trailing four calendar months. Fewer than 30 available traders produce fewer than 30 rows. Missing records never become zero trades. Public winner anecdotes and forum copy-trading claims suggest tests for concentration, specialization, latency and entry cost; they are not training labels or proof of a person's strategy.

Cost and integration rule

The self-built calculation engine, MCP server, stream fanout, job queue, FTS search and logger run on the existing stack. The dated gateway catalog lists Gemini 2.5 Flash-Lite at $0.10/M uncached input tokens and $0.40/M output; DeepSeek V4.1 Flash at $0.195/M and $0.78/M; GPT-5 nano at $0.05/M and $0.40/M. Provider availability, regional routing and tool quality still require live verification. Gemini remains the currently verified route. Model computation does not use an LLM.

Google AI Pro can include a claimed Google Developer Program cloud-credit benefit, and eligible ChatGPT Pro 100 subscriptions can activate API credits for a linked organization. These are account benefits, not credentials that the website can automatically borrow from a signed-in consumer chat session. No unverified credit balance is included in this app's operating forecast. Existing app and platform quotas remain active. The report distinguishes installed features, tested local experiments, live verification and future optional provisioning.

The bounded live Gateway comparison made exactly four requests: one scoped tool-routing step and one grounded follow-up per model. Both Gemini 2.5 Flash-Lite and GPT-5 nano passed both tasks. Gemini used 487 input and 123 output tokens, took 2.151 seconds across the two steps and cost $0.00009790; nano with minimal reasoning used 799 input and 179 output tokens, took 2.745 seconds and cost $0.00011155. Gateway-reported costs agree with the recorded token rates. Gemini remains the route because nano was 13.94% costlier and 27.58% slower in this tiny sample. This is not a general quality or p95 comparison and excludes remote MCP transport. `ai-gateway-benchmark.json` and the opt-in benchmark script preserve the exact requests, grading and costs.

The built-in probability-and-cost explanation now reads the scoped MCP packet directly, checks freshness and prints explicitly typed numerical facts with zero model calls. One hosted request completed in 1.36 seconds. General questions retain the low-cost assistant; contract quantities, side probabilities, dollar totals and per-contract costs are separate fields.

Ranking the requested integration surfaces

Priority	Surface	Benefit and implementation decision
1	Observability	Request traces, redacted structured logs, source-age checks, database diagnostics and usage allowances expose correctness and latency problems immediately. These are implemented in the existing processes first; additional paid retention or analytics products require a measured gap.
2	AI Gateway	The existing route supplies the read-only research assistant. A $3 monthly project budget, atomic application reservations and metered refunds bound usage; the actual two-model comparison above informs the current choice.

Priority	Surface	Benefit and implementation decision
3	Stores	Existing Supabase PostgreSQL and private object storage provide relational state, queues, full-text search and evidence archives. A second managed database/cache must beat the existing indexed/in-process path on the same workload before adding its bill and consistency work.
4	Marketplace wrappers	Provider-specific installation and secret provisioning can save integration work. Current Kalshi, ESPN, Spotify, X and MCP adapters already run in code; no additional paid wrapper was purchased. Compare a wrapper's exact capability, latency, rate limits and price against those adapters before adopting it.
5	Supabase OAuth integration	This guide builds an integration that other Supabase users authorize against their own projects. It becomes useful for a future multi-tenant product or external-project connector. The current app's user/private-journal boundaries and backend credentials are implemented directly; no broad account-management grant is introduced for this deployment.

Small predicted improvements remain experiment candidates. The partitioning experiment is an example of testing the hypothesis and keeping its negative result: the indexed heap was faster for that fixture. The row above records the present implementation and the evidence needed for a later replacement, rather than treating a product name as a performance result.

The existing single Railway backend was moved from SFO to Virginia without adding another service or volume. Seven pooled SELECT 1 samples per location measured mean database round-trip time of 55.30 ms before and 19.26 ms after, a 36.04 ms / 65.18% reduction in this short comparison. Applied configuration, the successful deployment manifest and a running instance confirm the destination region. This is a database-path measurement, not a whole-app speed claim. `deployment-latency.json` and `RUNTIME_RESEARCH.md` preserve the evidence and scope.

Live release checks exercised a $20-to-$30 scenario change, opening the matching paper ticket, a one-contract simulated buy, executable bid marking, an explicit risk-limit response and a close. The observed 18-cent entry and 15-cent net close produced a 3-cent realized paper loss, including modeled fees. Visible opinions and rankings refresh while the page is visible; expired sources, old quotes and failed refreshes become STALE and withhold old numerical projections. Decimal-second NBA clocks are parsed without confusing seconds and minutes; conflicting provider clock fields withhold an estimate. A real completed assistant call reserved $0.025, recorded a conservative $0.001 usage charge and refunded $0.024. Read-only hosted checks confirmed private-schema isolation and runtime grants after migration. These checks are engineering observations, not a trading-performance evaluation.

Rolling-deployment verification also exposed a six-connection runtime-role ceiling. It was raised to twelve for two overlapping instances, each with a four-connection pool and one listener, plus two operational sessions. Two consecutive six-client read-only probes passed without further backend growth; no role privileges, paid plan or pool size changed. Source outages now display a refresh failure instead of missing model coverage or raw internal errors.

Recruiter-facing claim wording

- Built a public prediction-market research and paper-trading application with shared WebSocket ingestion, bounded multiplexed portfolio updates, transactional PostgreSQL jobs, RLS, source lineage and immutable compressed evidence archives.

- Trained and compared ridge, XGBoost and neural NBA models on 2,569 real historical games; exported a calibrated ensemble to JavaScript with native-inference parity, chronological model selection and auditable residual-based sensitivity.

- Implemented fee- and depth-aware scenario sizing, discrete risk-constrained log-growth allocation, live paper-position decisions and confirmed settlement accounting; separated research estimates from realized paper results.

Use benchmark numbers only with their exact fixture and scope. Do not turn the 1.554% retrospective Brier point improvement into a general accuracy, profitability or return claim. No employment verification is part of this work.

Research sources and what they contribute

[1]-[6] supply earlier score-process, allocation, calibration and scoring methods. [7]-[11] describe research/engineering practice without disclosing proprietary strategies or certifying this app. [12]-[28] support earlier implementation/pricing decisions and public-trader boundaries. [29]-[34] supply context-filter and retrieval background. Later sources document protocol, source-access and platform boundaries. Inline links in each section identify the more specific supporting pages. Forum claims remain unverified and do not train outcome models.

The broad search included first-party research-firm material, original papers, platform documentation and public community discussion. The report does not infer a top trader's private tools or claim access to proprietary HRT methods. A direct official leaderboard request was throttled; no rate-limit bypass or private-profile retrieval was used.

[1] Stern: Brownian score-difference model

[2] Busseti, Ryu and Boyd: Risk-Constrained Kelly Gambling

[3] Kull, Silva Filho and Flach: Beta calibration

[4] Chen and Guestrin: XGBoost

[5] Guo et al.: On Calibration of Modern Neural Networks

[6] Gneiting and Raftery: Strictly Proper Scoring Rules

[7] HRT: Machine Learning and AI

[8] HRT: SharedWorker framework for real-time trading views

[9] HRT: Engineering, latency and research scale

[10] Two Sigma: Scientific research and technology approach

[11] Citadel: Alternative data research workflow

[12] PostgreSQL 17: Index types

[13] PostgreSQL 17: Partitioning

[14] W3C Trace Context

[15] Supabase Realtime source

[16] Supabase Broadcast

[17] Supabase Presence

[18] Supabase Postgres Changes

[19] Supabase Edge Function pricing

[20] Railway deployment regions

[21] Railway resource pricing

[22] Vercel AI Gateway live model catalog

[23] Google Developer Program credit benefits

[24] Eligible ChatGPT API-credit benefit

[25] Vercel: ChatGPT API credits integration

[26] Kalshi official leaderboard explanation

[27] Kalshi public leaderboard

[28] Community winner-sample discussion, unverified hypotheses

[29] Kalman (1960): A New Approach to Linear Filtering and Prediction Problems
[30] Rabiner (1989): A tutorial on hidden Markov models
[31] Howard et al.: Time-uniform Chernoff bounds via nonnegative supermartingales
[32] Supabase: selected-ticker hybrid search methods
[33] Hugging Face Transformers.js documentation
[34] MiniLM: deep self-attention distillation
[35] gRPC HTTP/2 protocol specification
[36] Protocol Buffers encoding guide
[37] Vercel Drains: current export pricing and schemas
[38] Supabase self-hosting scope and responsibilities
[39] Supabase Docker requirements
[40] PostgreSQL release versioning policy
[41] Supabase Wrappers: supported remote-system catalog
[42] X API pricing
[43] Reddit developer FAQ
[44] Rotten Tomatoes score definitions
[45] NBA official participation reports
[46] Google Ads API access migration
[47] Ploy company-owned LinkedIn page; identity remains tentative

Source-input manifest and reproducibility

The following SHA-256 values identify the source documents and numerical fixtures used for this render. They identify inputs, not proof of production deployment. Full source, exact test commands and capture limitations remain in the repository.

Input	SHA-256
web\docs\FEATURE_REVIEW.md	48bff4f75ef20fd3bf3ea8a84ecd1b16513511ff4aaf8cf4d34013a7095e2584
web\docs\EVIDENCE_RESEARCH.md	92bf14a9a71597635ebfb9d684162138b853d1228676f8d0912af8c8c5f4ee5b
web\docs\QUANT_CONTEXT_EXPERIMENTS.md	e52e62c0ac0827573007530ca67a7544b483e4dccbc7107c8affc43385a311fa
web\docs\SEMANTIC_EVIDENCE.md	17e1cc4b5c8b355ad384f2bac0a0b00cc3008745ebfd4ab90052797d520c5bdf
docs\account-activation.md	ed5156940927d120d63326a263b65e2db67dbb2363631f3e14213b139557b33a
web\docs\ANALYTICS_SETUP.md	e49ca9fddd5a88be3e41cbb0397b4ab56afdb80dffc071f2942fff9b02df08ec
docs\engineering-runtime.md	8596ebdfde36c9caaca6c2e90c4b58c03da07817d1914b4bca04cff10610e017
docs\request-matrix-v3.md	9b97865c03909fba93214966cd8696dff7ae60bdba8c3c0f4d4d4558371692d5
web\docs\V3_VERIFICATION.md	a2021415a8cef22f8c646d8fb2a42335c3ec1024b793ec78c70c5a1d968fa8c1
web\docs\nba-challenger-validation.json	06ad545b35251c0a6085919c297bddcdc541eda703dbb5b33ab71acf143f74e6
web\docs\nba-inplay-validation.json	f51a72de8f29142be74caa63d5ade2165917e09720e1ed3e2a1fd4f25cacd7cc
web\docs\runtime-benchmark.json	dc402ad2c5c6a8cfd0e29ee08918a85356941934b9b4bf3aae4f803a377c05a1
web\docs\sizing-benchmark.json	6d0e879953736d27eb7195e7517d3db5f35e5a5e1f75143dd5186b2db6da4517
web\docs\nba-inference-benchmark.json	66b46d26ec292cd4c0cbb311d4ee5d6976787136463ebb2402d33ee91591c091

Input	SHA-256
web\docs\semantic-verification.json	15331a31b29a80916618710b5c43f63877c44fe1ffbeaeb1749ee0487f073083
web\docs\evidence-verification.json	9b3c265dbebde9eab75101215d3abaa2e82ec85dd6a0608570404f7297495262
review\browser-validation-v3.json	aa32f2da490bf177e94f4b970f0de991cd263e55c7fc1efaf4234ee212338ee7
review\npm-test-after-account-fixes.txt	a6380bb338ddf4f83d71827bcc054c76c8d20126fc2c26a6d358d93aa9b134bb
review\typecheck-final.txt	65d3ab07af05689af3b39ce121bce028ba0447deb8b4eec4f42e4bee3835ed3a
review\next-build-final.txt	e0042395f2f593f8abed3ac6e2f664a233b892090623d4c46ff412f6b85e5bdc